

UNO-PI

AI-Powered Underground Mine Safety, Monitoring and Rescue System

1. System Architecture Overview

ResQtron is built on a two-platform architecture: a ground rescue rover as the primary unit and an aerial drone as a secondary support unit, both reporting to a central base station operated by the rescue team. The overall system follows a closed-loop operational cycle — Sense → Process → Think → Act → Report — in which environmental data is continuously collected, analysed, and acted upon before being relayed to human operators for verification and decision-making.

The rover's ESP32 controller acts as the central processing unit, handling sensor polling, decision logic, motor actuation, camera streaming, and LoRa communication concurrently through an RTOS-based multitasking scheduler. This ensures that critical functions such as gas monitoring and obstacle detection continue uninterrupted even while the rover is navigating or transmitting data.

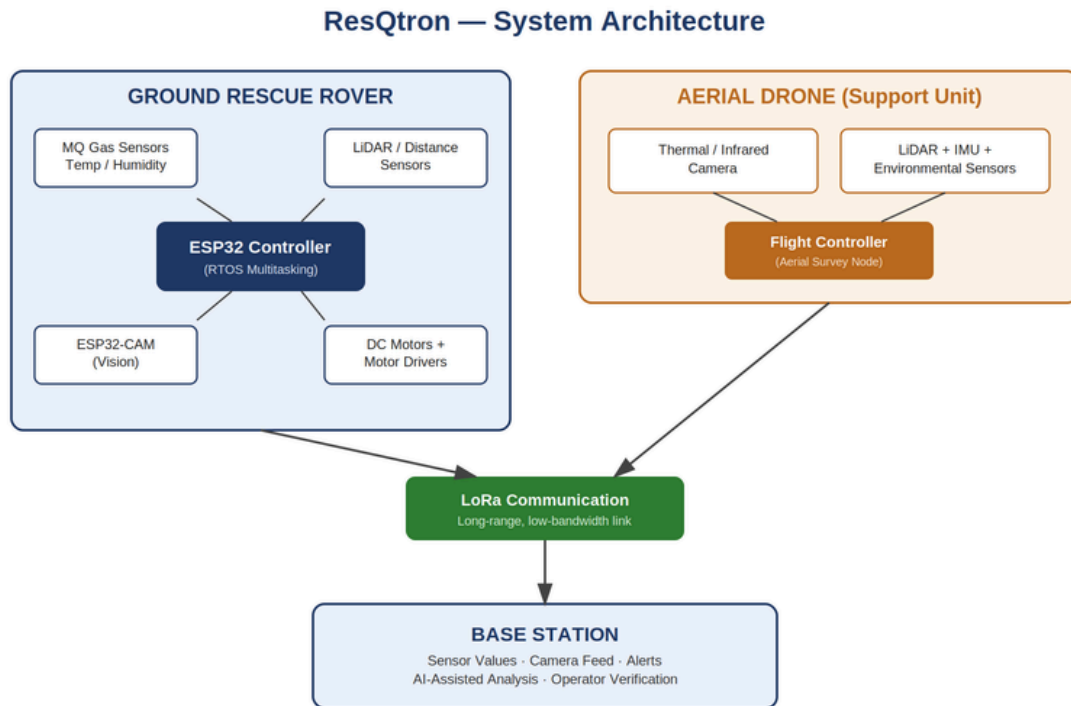


Figure 1: ResQtron System Architecture — Rover, Drone, LoRa link and Base Station

2. Component Selection and Justification

2.1 Controller — ESP32

The ESP32 was selected as the core embedded platform because of its low cost, built-in Wi-Fi/BLE connectivity, sufficient GPIO count for multiple simultaneous sensor and actuator connections, and native support for RTOS-based multitasking. This allows the controller to run several operations — sensing, navigation, communication, and alert handling — in parallel rather than sequentially, which is essential in a time-critical rescue scenario.

2.2 Gas Sensing — MQ-Series Sensors

MQ-series gas sensors were chosen for their ability to detect toxic and flammable gases commonly present in underground mine environments, such as methane and carbon monoxide. They are low-cost, widely available, and well-documented, making them suitable for rapid prototyping within the 24-hour build window while still providing meaningful hazard detection.

2.3 Environmental Sensing — Temperature and Humidity Sensors

Temperature and humidity sensors continuously track environmental conditions inside the mine, allowing the system to flag rising heat levels or unstable atmospheric conditions that could indicate fire risk or structural instability.

2.4 Navigation and Obstacle Sensing — LiDAR/Distance Sensors

LiDAR or distance sensors were selected over simpler ultrasonic alternatives for their higher accuracy and range in detecting nearby obstacles, blocked pathways, and changes in tunnel geometry, enabling more reliable navigation decisions in confined, irregular mine passages.

2.5 Vision — ESP32-CAM

The ESP32-CAM module provides a real-time visual feed to the operator at the base station, supporting both manual visual inspection and AI-assisted analysis for identifying obstacles or possible trapped workers.

2.6 Mobility — DC Motors with Motor Drivers

DC motors paired with motor drivers give the rover forward, reverse, turning, and stopping control, offering a simple and reliable drivetrain that can be assembled and tuned quickly during the hackathon while remaining robust enough for rough mine-floor terrain.

2.7 Communication — LoRa Module

LoRa was selected over Wi-Fi or Bluetooth specifically for its long transmission range and reliable performance in low-bandwidth, obstruction-heavy environments such as underground tunnels, where conventional wireless protocols would fail. This ensures that critical alerts and sensor readings still reach the base station even from deep or partially blocked sections of the mine.

2.8 Drone Payload

The supporting drone carries a thermal/infrared camera, LiDAR, IMU, and environmental sensors, enabling rapid aerial surveying and heat-signature-based detection of possible worker presence in low-visibility areas that the ground rover cannot easily access.

3. Engineering Design Rationale

The system is designed as a fully modular architecture, with the sensing module, controller, communication module, motor-control section, power system, camera system, and mechanical mechanisms all separable and independently testable. This modularity reduces integration risk during the 24-hour build, since each subsystem can be validated on its own before being combined into the complete prototype.

RTOS-based multitasking on the ESP32 is central to the design philosophy: because gas monitoring, navigation, camera streaming, communication, and alert handling must all run simultaneously during an active rescue, a single-threaded control loop would be inadequate. The RTOS scheduler allocates dedicated tasks to each function, ensuring no single operation blocks another.

4. Technical Feasibility

ResQtron uses commercially available, low-cost, and well-documented components — ESP32, ESP32-CAM, MQ-series gas sensors, LiDAR modules, and LoRa transceivers — all of which have mature libraries and community support, making them realistic to prototype and debug within a 24-hour hardware hackathon timeframe.

LoRa's long-range, low-power communication profile is technically well-suited to underground and obstructed environments where Wi-Fi or cellular connectivity would typically fail, directly supporting the project's core requirement of reliable data transmission from hazardous zones back to a safe base station.

5. Hardware–Software Integration

The ESP32 firmware integrates sensor polling, motor control, camera streaming, and LoRa packet transmission under a single RTOS scheduler, coordinating all rover functions from one embedded platform. At the base station, an AI-assisted analysis layer processes incoming camera and sensor data to flag probable hazards or worker detections, which are then presented to the human operator for final verification — keeping trained personnel in the loop for all critical decisions.

Summary Table — Key Engineering Choices

Subsystem	Component	Reason for Selection
Controller	ESP32 (RTOS)	Low cost, multitasking, sufficient GPIO
Gas Sensing	MQ-series sensors	Detects toxic/flammable gases, low cost

Environment	Temperature & humidity sensors	Tracks heat rise and unstable conditions
Navigation	LiDAR/distance sensors	Accurate obstacle and path detection
Vision	ESP32-CAM	Real-time visual feed for operator/AI
Mobility	DC motors + motor drivers	Simple, robust drivetrain control
Communication	LoRa module	Long range, reliable in obstructed terrain
Drone Payload	Thermal cam, LiDAR, IMU	Aerial survey, heat-signature detection